

Beyond Outbidding: Armed Group Responses to Transnational Jihadist Competition*

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Abstract

Numerous jihadist organizations pledged allegiance to al-Qaeda (AQ) or the Islamic State (IS) in recent years. How do armed groups respond to competition from transnational jihadist rivals? A prominent strand of research argues that increased competition leads armed groups to *outbid* rivals through increased violence. However, alternative theories suggest that groups can *restrain* their violence to distinguish themselves from extremist rivals and appeal to local and international audiences. We test these competing hypotheses surrounding outbidding and restraint using original data on pledges to AQ and IS and a Difference-in-Differences (DiD) analysis of militant violence against civilians. We find that pledges to AQ or IS are associated with a moderate decline in the violence of other armed groups against civilians. Importantly, these findings are primarily driven by non-religious organizations and groups that are not formally designated as terrorists. This study contributes to research on militant competition, civilian victimization, and transnational actors.

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1 Introduction

Dozens of jihadist organizations pledged allegiance to al-Qaeda (AQ) or the Islamic State (IS) over recent decades. For AQ, this included jihadist groups like Jama’at al-Tawhid wal-Jihad in Iraq, the Salafist Group for Preaching and Combat in Algeria, and al-Shabaab in Somalia. After declaring its caliphate in June 2014, IS also received pledges of allegiance from numerous jihadist groups, including Ansar Bayt al-Maqdis in Egypt, Boko Haram in Nigeria, Ahlu Sunna wal Jama’a in Mozambique, and the Bangsamoro Islamic Freedom Fighters in the Philippines (Jadoon, 2018; Warner et al., 2022; Krause, 2024). Participating in insurgencies throughout Africa, Asia, and the Middle East, AQ and IS “affiliates” compete against other armed groups for scarce resources and support.¹ How do armed groups respond to competition with these transnational jihadist rivals?

Competition between armed groups can shape the trajectory of organizational violence (Kydd and Walter, 2006; Nemeth, 2014; Findley and Young, 2012; Phillips, 2015; Conrad and Spaniel, 2021; Polo and Welsh, 2024). A central theory in conflict research suggests that increased competition in violent political markets encourages armed groups to *outbid* rivals through violent escalation (Bloom, 2004; Kydd and Walter, 2006; Conrad and Greene, 2015). By increasing the scale and scope of their violence, armed groups generate publicity, demonstrate resolve, and show their capacity to achieve organizational goals. In contrast, another body of research argues that armed groups also have incentives to strategically *restrain* their violence (Stanton, 2013; Jo, 2015; Stanton, 2016; Fazal, 2018; McQuinn et al., 2021). Adopting a

¹Recent research defines affiliates as organizations “that openly support either IS or al-Qaida, are supported by either IS or al-Qaida, or have pledged allegiance to one of the groups but are still waiting for a formal acceptance” (Pettersson and Öberg, 2020, 603). However, given that pledging allegiance differs from other forms of support and is required for acceptance into AQ and IS, we conceptualize affiliates as jihadist groups that have pledged fealty (*bay’a*) to AQ or IS.

strategy of restraint during competition can allow armed groups to attain legitimacy by differentiating themselves from extremist rivals, highlighting rules of engagement that avoid civilian harm, and signaling adherence to international law and norms surrounding appropriate wartime conduct.

To test these competing theories, we analyze novel data on jihadist pledges of allegiance to AQ and IS. The data on pledges were compiled by examining Arabic- and English-language sources on the organizational histories of 161 jihadist groups operating in 41 countries to determine whether each group pledged allegiance to AQ or IS.² Using Difference-in-Differences (DiD) models, we analyze a time-series cross-sectional dataset that examines the behavior of 485 armed groups from 2004 to 2016. This list of 485 groups includes all organizations, both religious and non-religious, that remained unpledged to AQ or IS during this time period. Specifically, we focus on these groups' violence against civilians. By using the emergence of pledges to AQ or IS as a quasi-experimental treatment, our design seeks to isolate the causal effects of transnational jihadist competition on the use of civilian victimization by other (unpledged) armed groups in a conflict.

Our baseline difference-in-differences analysis suggests that armed groups tend to respond to transnational jihadist competition with *restraint* rather than violent *outbidding*. We find that pledges to AQ or IS affiliates in a conflict are associated with a moderate decline in civilian victimization by other armed groups—about a 16 percent reduction in attacks against civilians; however, this reduction is only significant at the 90 percent confidence level. At the same time, the results do not indicate a comparable decrease in organizational attacks against military or police forces. This suggests that the change in group behavior may reflect a deliberate adjustment

²Jihadist organizations constitute Islamist groups that justify the use of violence to achieve religiously-based goals (Ahmad, 2019).

in civilian targeting rather than a general decrease in organizational violence. Yet, these effects are not uniform across all unpledged groups and are primarily driven by non-religious and non-terrorist designated groups, where reductions in civilian victimization are larger and reach statistical significance at the conventional 95 percent level in some models. There also are temporal dynamics associated with these findings, with restraint occurring several years after the appearance of pledges in a country. Finally, much of this restraint effect documented among organizations that remained unpledged to AQ/IS is driven by pledges to IS, rather than to AQ, that occurred in their country of operation.

This study contributes to multiple areas of conflict research. Contemporary intrastate conflicts are increasingly defined by fragmentation and the presence of numerous armed groups (Malone, 2022; Lewis, 2023), with modern conflict being “more complex, more fragmented, and harder to define than ever before” (ACLED, 2025). Focusing on multi-party conflicts, our approach emphasizes how competition with different *types* of actors—in this case, transnational jihadist groups—affects incentives among different militant organizations to engage in outbidding or restraint. Methodologically, we contribute to the literature on militant competition by conducting one of the first DiD tests in this context.

Second, scholars have explored the conditions under which armed groups limit violence against civilians (Balcells and Stanton, 2021; Downes and Rangazas, 2023). In doing so, key findings have shed light on the importance of factors such as organizational ideology (Revkin and Wood, 2021), embeddedness in local communities (McQuinn et al., 2021), territorial control (Kalyvas, 2006), and capabilities (Wood, 2010). While existing literature often views competition as a driver of escalation, our findings highlight that competition can lead groups to restrain their violence. These findings contribute to recent literature surrounding armed groups’ efforts to generate

legitimacy—“the belief in the rightfulness of an armed group’s agenda and violent struggle” (Schlichte and Schneckener, 2015, 413)—in conflicts around the world.

Finally, the jihadist movement has expanded dramatically in recent decades (Walter, 2017*b*). Among jihadist groups, AQ, IS, and their affiliates have played an active role in numerous conflicts; in 2019 alone, these transnational organizations participated in more than half of all civil conflicts in the international system (Pettersson and Öberg, 2020). To date, scholars have examined why jihadist groups cooperate with AQ and IS and how such links can affect the behavior of pledging groups (Farrell, 2020; Blair and Potter, 2023). Yet, relatively less scholarship has explored how rhetorical cooperation with AQ and IS affects the behavior of unpledged groups operating in the same conflict. Understanding these dynamics remains a pressing concern for scholars and policymakers, as the rise and “expansion of transnational jihadist groups like IS and al-Qaida and their affiliates represents one of the most distinctive trends in organized violence since the end of the Cold War” (Melandar, Pettersson and Themnér, 2016, 731).

2 Pledging Allegiance to al-Qaeda and the Islamic State

Leaders of jihadist groups operating across multiple regions have pledged allegiance (*bay’a*) to AQ and IS. This pledge carries salient historical and religious connotations. Early members of the Muslim community pledged *bay’a* to the Prophet Muhammad, and oaths of allegiance were subsequently given to other leaders in the Islamic world, such as caliphs (Milton and Al-Ubaydi, 2015; Wagemakers, 2015). Rather than mere rhetorical support, pledges formed a binding agreement: “scholars saw violating the *bay’a* as punishable by death, unless the caliph did not live up to his duty to uphold

the rules and regulations of Islam, in which case those who pledged allegiance were freed from their obligations towards the ruler” (Wagemakers, 2015, 99). In modern times, state leaders, such as Muammar Gaddafi and Saddam Hussein, adopted the tradition of receiving *bay’a* to legitimize their rule. This means that authoritarian and non-religious leaders may have altered perceptions surrounding the meaning of pledging allegiance (Porter, 2015).

However, given their Salafist identity and the desire to emulate the behavior of the early Muslim community, oaths of allegiance carry significant meaning for AQ, IS, and pledging groups (Wagemakers, 2015). Pledging allegiance is a necessary condition for jihadist leaders and their groups to gain formal acceptance into the ranks of AQ or IS (Blair et al., 2022; Krause, 2024). Moreover, offering one’s allegiance to AQ or IS publicly recognizes their authority and legitimizes their status claims. Coming under the authority of these transnational actors, pledges also create a contract; breaking this oath constitutes a major breach that could significantly damage an actor’s standing in the jihadist movement (Bacon and Arsenault, 2019).

By pledging allegiance to these transnational organizations, jihadist groups thus make a strategic decision that signals their commitment to the ideologies, military strategies, and tactics promoted by AQ or IS (Brzezinski, 2024). If these pledges are recognized by the leadership of AQ or IS, pledging groups may assume responsibility for operations in entire geographic regions, such as the “Islamic Maghreb” or “Greater Sahara.” In many cases, pledging groups espouse new identities and abandon previous names in favor of AQ or IS branding (Byman, 2014). Despite notable exceptions, such as Jabhat al-Nusra, pledges have largely served as the basis for long-term strategic partnerships between AQ, IS, and pledging groups that have endured years of state repression and leadership decapitation (Moghadam, 2017).

Through this expansion strategy, AQ and IS have influenced conflict dynamics in

multiple regions. AQ, IS, and their affiliates accounted for more than 70 percent of global fatalities caused by all armed groups in one-sided violence in 2019 and caused the majority of battle-related deaths in seven out of 10 years between 2010 and 2019 (Pettersson and Öberg, 2020). Beyond targeting local civilians and regional regimes, the AQ and IS networks have launched attacks in Europe and the United States as part of their efforts to combat “Crusader” and “Imperialist” powers. Being among the most lethal armed groups in the international system, AQ, IS, and their affiliates developed reputations for engaging in mass casualty attacks and indiscriminate violence.

For one, AQ played a key role in the global diffusion of suicide attacks (Moghadam, 2009). AQ affiliates have also frequently used this tactic. In 2007, the Salafist Group for Preaching and Combat, which changed its name to al-Qaeda in the Islamic Maghreb, executed its first suicide attack just months after pledging fealty to AQ. Al-Qaeda in Iraq, for its part, accounted for a disproportionate percentage of deaths caused by armed groups in Iraq through its use of suicide attacks and indiscriminate violence against civilians (Piazza, 2009). In another case, al-Shabaab has conducted dozens of suicide attacks, killing more than 1,200 civilians in suicide bombings between 2011 and 2019 (Shire, 2023). Al-Qaeda in the Arabian Peninsula also executed numerous suicide attacks in Yemen, including a 2015 suicide bombing attack in central Sana’a that killed at least 47 people (Reuters, 2014).

Additionally, IS attracted worldwide attention for its rapid territorial expansion and large-scale violence. This involved massacres, sexual violence, videotaped executions, beheadings, the burning and drowning of hostages, and crushing captives with tanks (Friis, 2018). Jihadist groups that pledged their allegiance to IS have often emulated IS’s brutality and violent tactics (Brzezinski, 2024). Ansar Bayt al-Maqdis, which pledged to IS in 2014, claimed responsibility for the 2015 downing of

Metrojet Flight 9268, an attack that killed more than 220 passengers. Elsewhere, the Islamic State Khorasan’s 2021 suicide bombing at the Kabul Airport, which killed 170 civilians and 13 U.S. soldiers, is noted to have “sent shockwaves around the world” (Jadoon et al., 2024).

This discussion is not intended to suggest that AQ and IS are monolithic entities or that these groups do not place restrictions on their violence. AQ leaders, for example, have called for greater restraint and moderation in recent years (Hamming, 2020). At the same time, AQ, IS, and their affiliates have global reputations for conducting highly-lethal acts of violence. Due to such histories, states and international actors devote significant resources to combating these transnational jihadist groups and often formally designate them as terrorists (El Masri and Phillips, 2024). Consequently, AQ, IS, and their affiliates are often viewed as more extreme relative to other armed groups. In the following sections, we explore how and why competition with these transnational jihadist groups may affect the behavior of unpledged groups operating in the same conflict.

3 Competition and Escalation in the Shadow of Transnational Jihadists

Armed opposition movements are frequently fragmented and contain dozens, if not hundreds, of organizations. Such competitive settings force armed groups to compete for scarce resources and support (Balcells and Stanton, 2021). However, civilians are often uncertain about the true intentions and capabilities of rival groups (Kydd and Walter, 2006). To overcome this uncertainty, armed groups can calibrate their violent behavior, selecting from a variety of violent strategies to project strength, generate publicity, and signal commitment to their cause.

In fragmented conflicts, one prominent strategy that armed groups may pursue is *outbidding*. Violent outbidding involves engaging in higher levels of violence and the use of increasingly innovative and severe tactics (Farrell, 2020). Distinguishing themselves from rivals through increased violence, armed groups can signal resolve and effectiveness to potential supporters (Kydd and Walter, 2006; Conrad and Spaniel, 2021). “While committing more shocking attacks” may not necessarily help an organization achieve its long-term goals, Conrad and Greene (2015, 549) write that “the spectacle of more severe attacks might generate the *perception* that the organization is more effective than it actually is.” By creating such a perception, armed groups can increase their “market share” of popular support at the expense of rivals (Bloom, 2004).

The outbidding model of violence—described as “one of the most widely cited theories of [armed] groups’ motivations and actions” (Farrell, 2020, 437)—has been used to explain the use of various tactics and targeting strategies of armed groups. For example, scholars have drawn on the logic surrounding outbidding to explain attacks against NGOs, educational institutions, and U.S. citizens in foreign countries (Murdie and Stapley, 2014; Asal, Phillips and Rethemeyer, 2022; Asal et al., 2023). Competition between armed groups has also been linked to the increased use of indiscriminate tactics, such as suicide bombings, and higher levels of civilian victimization (Bloom, 2005; Wood and Kathman, 2015; Farrell, 2020). Importantly, escalating violence against civilians can occur among intrafield rivals — or between groups with similar goals — as well as between groups with different motivations seeking to represent separate populations (interfield rivals) in a conflict zone (Conrad, Greene and Phillips, 2023).

In competitive settings, armed groups have various incentives to target civilians. First, attacks against civilians can be easier to carry out than operations against

military or police targets. This means that groups that seek to increase the quantity of their attacks and stand out in competitive, low information settings may focus on soft targets (Asal, Phillips and Rethemeyer, 2022). Moreover, armed groups can employ violence to ensure civilian compliance and deter defections to rivals (Kalyvas, 2006; Wood and Kathman, 2015). Violence against civilians can further demonstrate a group’s commitment to achieving its political goals: “By engaging in what is often perceived to be the most extreme form of violence, the faction attempts to win support and notoriety by drawing attention to itself and framing itself as the group most committed to the cause” (Belgioioso and Thurber, 2024, 335).

Based on these dynamics, competition from transnational jihadist organizations can form a unique opportunity for unpledged groups operating in the same conflict to generate publicity, demonstrate their resolve, and highlight their ability to achieve organizational goals. After all, AQ, IS, and their affiliates carry international reputations for extremism, using mass violence against civilians, and refusing to negotiate with state and international actors (Walter, 2017*a*). Facing competition from such actors may thus lead unpledged groups to respond by increasing their levels of violence against civilians in their attempts to outbid these transnational jihadist groups.

Hypothesis 1. Unpledged armed groups facing competition from a transnational jihadist rival will *increase* their violence against civilians.

Despite its prominence in conflict research, empirical tests of outbidding’s theoretical expectations have yielded mixed results (Findley and Young, 2012; Crisman-Cox and Gibilisco, 2024). In light of these uncertain findings, an expanding body of literature examines how armed groups may deliberately restrain their violence, even when they possess the ability to do otherwise.

4 Strategic Restraint During Competition

In this section, we theorize how competition with transnational jihadist groups can incentivize unpledged armed groups operating in the same conflict to restrain their violence against civilians. Busher and Bjørge (2020, 3) define *restraint* as a conscious process in which “militants *choose* to drop, downscale, or limit an attack or campaign, or adopt tactical or strategic innovations that lead them away from violence.” Such decisions reflect the recognition that certain forms of violence may be strategically counterproductive to securing popular support, misaligned with group identities, or harmful to organizational survival (Busher and Bjørge, 2020; Macklin, 2020; Onder, 2024). Restrained violence can thus be an important aspect of an organization’s legitimacy-seeking efforts, or “the range of behaviors used by a rebel group to engender non-coercive compliance and adherence to their authority that is directed toward diverse audiences irrespective of the ultimate success of such behaviors” (McWeeney, Cunningham and Bauer, 2025, 270). Specifically, we outline how armed groups can use restraint to attain legitimacy among local and international audiences by differentiating themselves from extremist rivals, showing their concern with avoiding unnecessary harm to civilians, and underscoring their commitment to international norms and law.

4.1 Restrained Competition

Armed groups engage in various behaviors to attain and maintain legitimacy among different audiences. Seeking legitimacy among local populations, armed groups have created participatory institutions and provided various public goods and services, such as healthcare, education, and dispute arbitration (Arjona, Kasfir and Mampilly, 2015; Stewart, 2018; Cunningham, Huang and Sawyer, 2021; Breslawski, 2023). In

doing so, armed groups can “out-legitimize” other actors, showcasing their capacity to govern more effectively than rivals (Furlan, 2023). Public outreach campaigns and diplomatic efforts also play an important role in organizations’ attempts to secure legitimacy among international audiences, such as great powers, regional states, and international governmental organizations (Coggins, 2015; Kim and Onder, 2025).

In addition to such tactics and strategies, exercising restraint can serve as a deliberate choice aimed at securing legitimacy (Stanton, 2016; Garbino, 2023). After all, despite its potential to generate publicity and demonstrate resolve, excessive violence can delegitimize armed groups among local populations, especially when it targets civilians. As Polo (2020, 236) writes: “Terrorist attacks against undefended civilians...can be much more costly than attacks on official targets in terms of loss of popular support and legitimacy.”

By restraining their violence against civilians, armed groups can highlight clear rules of engagement that set standards for future behavior (Schlichte and Schneckener, 2015). After all, it is critical for armed actors to ensure that local populations perceive them to be engaged in just conduct that avoids unnecessary civilian harm (Krick, Petkun and Revkin, 2025). The use of restraint also allows armed groups to differentiate themselves from extremist rivals in competitive environments. Differentiation denotes “a process by which actors move away from the position of others in a social system through communication and/or behavior” (Schwab, 2023, 5). By limiting their violence against civilians, armed groups can portray themselves as relatively moderate organizations compared to rivals, helping to maintain legitimacy among local populations.

Armed groups may also attempt to cultivate legitimacy on the international stage through restraint. Militant actors often face legitimacy deficits in the international community due to their commitment to using violence to achieve organizational

goals (Podder, 2017). Involvement in terrorism can further decrease armed groups' support among external audiences (Arves, Cunningham and McCulloch, 2019) and subject organizations to international condemnation and blacklisting (Jo, Phillips and Alley, 2020). To avoid such issues, armed groups can signal compliance with international norms and law surrounding appropriate forms of wartime behavior (Jo, 2015; Lasley and Thyne, 2015; Downes and Rangazas, 2023). Fazal (2018, 197) states that by refraining from targeting civilians, armed groups can "send signals to the international community of their willingness and capacity to be good citizens of that community." Rather than being a short-term strategy, restraint can serve as a hand-tying mechanism for future group behavior (Loyle et al., 2023), as reverting to previous levels of civilian victimization can be costly among international as well as domestic audiences.

Together, these insights highlight how restraint offers unpledged armed groups operating in a conflict where a jihadist rival has pledged to AQ or IS an alternative pathway to outbidding. Transnational jihadist rivals can escalate violence to extreme levels in competitive violent markets. This leaves little room for escalation to occur without alienating local populations or provoking international backlash. In contrast, armed groups can draw on restraint as a deliberate strategy in conflicts. Limiting violence against civilians allows armed groups to differentiate themselves from more extreme rivals, outline rules of engagement that avoid unnecessary civilian victimization, and signal commitment to international law. In doing so, armed groups can achieve and preserve legitimacy among local and international actors.

Hypothesis 2. Unpledged armed groups facing competition from a transnational jihadist rival will *reduce* their violence against civilians.

4.2 Ideological Distance and the Incentives for Restraint

However, incentives to engage in restraint may not be uniform across all armed groups. Instead, these incentives may vary according to a group’s ideological distance from transnational jihadist rivals. Ideology denotes a set of ideas that identify a constituency and outline grievances, specific goals, and a program of action to achieve organizational objectives (Sanín and Wood, 2014). Within violent political markets, ideology functions as both a guiding framework for action and a symbolic marker that distinguishes armed groups from their competitors (Akcinaroglu et al., 2025).

Armed groups that share ideological affinities with other militant organizations can struggle to distinguish themselves from these rivals. For example, while Ahrar al-Sham in Syria successfully differentiated itself from jihadist groups like IS and Jabhat al-Nusra, it failed to construct meaningful differences with Hay’at Tahrir al-Sham, making the two Islamist groups less distinguishable among local populations and international actors (Schwab, 2023). The ability of some armed groups to differentiate themselves from rivals to external audiences may also be constrained. For example, some U.S. policymakers have viewed “radical Islam” as a monolithic movement for years, seeing little distinction between disparate Islamist groups despite their adoption of significantly different goals, strategies, and tactics (Lynch, 2017). Due to these difficulties, armed groups with greater ideological distance from transnational jihadist groups may be able to more effectively leverage restraint to highlight their differences from these organizations.

In sum, ideological distance from transnational jihadists can amplify the signal of violent restraint. This helps unpledged armed groups signal not merely a short-term tactical adjustment, but rather a principled commitment to behavioral restraint. Unlike unpledged armed groups with overlapping religious narratives that may struggle

to convince observers of their meaningful differences from transnational jihadists, ideologically distant unpledged groups can leverage restraint to project a moderate image of organizational behavior and goals.

Hypothesis 3. Unpledged armed groups that are *ideologically distant* from transnational jihadist rivals will *reduce* violence against civilians *more* than unpledged groups that share ideological affinities with these organizations.

4.3 Terrorist Designations and the Incentives for Restraint

Although ideological distance from transnational jihadists can enhance the credibility of restraint as a signal, not all unpledged groups are equally positioned to translate this credibility into reputational gains, particularly on the international stage. As a result, an unpledged armed group’s reputation can further shape the incentives for adopting restraint as a competitive strategy. Among different indicators, terrorist designations stand out as particularly powerful markers of an armed group’s standing in the international community.

The “terrorist” label codifies perceptions of organizations as illegitimate actors in the international system. Once a group is labeled a terrorist organization, it becomes subject to economic sanctions, asset freezes, and travel restrictions, limiting its political and operational flexibility (Jo, Phillips and Alley, 2020). This reputational damage can be irreparable given the long-lasting and “sticky” nature of the terrorist label (Phillips, 2019). Terrorist designations can thus carry far-reaching consequences that can blunt incentives for restraint.

Unpledged armed groups that have avoided terrorist designation might be better positioned to leverage strategic restraint as a tool for maintaining their international

appeal in the wake of competition. Their relatively untainted reputations leave open the possibility of building legitimacy in the eyes of international actors. In contrast, unpledged groups already burdened by terrorist designations face significant constraints in translating restraint into reputational gains and may have little incentive to seek international legitimacy. As a result, an unpledged armed group’s pre-existing reputation and prospects to reshape external perceptions may affect the incentives for pursuing restraint during competition with transnational jihadist rivals.

Hypothesis 4. Unpledged armed groups *without prior terrorist designations* will *reduce* violence against civilians *more* than unpledged armed groups already designated as terrorists.

Before proceeding, it is important to note a number of scope conditions for these arguments. This paper analyzes the effects of competition with AQ, IS, and their affiliates on the behavior of unpledged armed groups operating in the same conflict. Consequently, this study does not seek to explain the competitive dynamics of conflicts that do not involve these transnational jihadist organizations. Additionally, the growing number of jihadist groups and the increasing transnationalization of jihadist organizations’ demands are two key defining traits of post-2003 civil wars (Walter, 2017b). Although this study helps to understand contemporary conflict dynamics, competition with AQ and IS is a recent phenomenon, limiting understanding of pre-2003 conflicts.

Additionally, numerous conflict zones witness competition between organizations that espouse similar ideologies as well as among rivals with significantly different belief systems. Moreover, designated terrorist organizations operate in numerous countries around the world; since 2015, “a clear majority of internationalized intrastate conflicts” involve formally designated terrorist groups as major conflict ac-

tors (Schissler et al., 2026, 21). While we focus on competition with AQ and IS, our study’s theoretical discussion and findings might still travel to other fragmented conflicts involving formally designated terrorist groups.

5 Research Design

We exploit pledges of allegiance given by jihadist organizations to AQ and IS as a quasi-experimental treatment within a Difference-in-Differences (DiD) framework. This allows us to assess how unpledged armed groups adapt their behavior in response to the onset of competition from transnational jihadists.

Our DiD models are based on a time-series cross-sectional dataset. Specifically, we analyze the behavior of 485 armed groups from 2004 to 2016 using the Extended Data on Terrorist Groups (EDTG) dataset (Hou, Gaibullov and Sandler, 2020). These are armed organizations possessing disparate ideologies that remained *unpledged* to AQ or IS (“non-AQ/IS-affiliated groups”). The EDTG dataset provides broad coverage of armed group behavior by incorporating organizations listed in the Global Terrorism Database (GTD), regardless of their violence levels. This inclusive approach aligns with our objective of examining whether various types of armed groups increase or reduce violence in response to competition with AQ and IS.

Our primary outcome of interest is violence against civilians conducted by the 485 armed groups that did not pledge allegiance to AQ or IS. We examine changes in these groups’ violence against civilians following pledges of allegiance to AQ or IS by jihadist groups operating in the same country. The analysis begins in 2004, the year of the first pledge to AQ by a non-affiliated group (Byman, 2014; Mendelsohn, 2015). Pledging remained relatively rare throughout the 2000s, allowing us to establish a clear behavioral baseline for armed groups.

5.1 Civilian Victimization

We measure civilian victimization using incident-level data from the GTD, which categorizes 22 target/victim types. We classify attacks as civilian victimization if the GTD lists targets as private citizens, businesses, educational institutions, journalists, religious figures, tourists, NGOs, airports, maritime, telecommunications, transportation systems, or abortion clinics.³ Our primary measure, *Civilian Victimization*, counts the number of such attacks by a non-AQ/IS-affiliated group in a given year.⁴ We also conduct additional tests using narrower categories of civilians in Appendix 16 and report robustness checks using *Number of Civilian Casualties* and negative binomial models with the count of attacks (Appendix 4 and 12).

5.2 Measuring Pledges to al-Qaeda and the Islamic State

We collected data on pledges to AQ and IS by analyzing the organizational histories of 161 jihadist groups that were active at some point after 2004 in 41 countries across Africa, Asia, and the Middle East. These were militant Sunni groups that could have potentially pledged allegiance to either AQ or IS, as other types of armed groups are not eligible to become AQ or IS affiliates. We created a list of jihadist organizations by reviewing armed groups in the Armed Group Dataset (Malone, 2022), Non-State Actors in Armed Conflict Dataset (Cunningham, Gleditsch and Salehyan, 2013), Big, Allied and Dangerous (Asal and Rethemeyer, 2015), End-of-Terror Dataset (Jones and Libicki, 2008), and the Mapping Militants Project (Crenshaw and Robinson, 2025).

For each jihadist group, we identified whether a formal pledge of allegiance had

³This excludes attacks on government, police, military, and violent organizations.

⁴We log this variable after adding 1 to address zero values, following standard practice.

been given to AQ or IS. In doing so, we omit cases of broad ideological support, since formal pledges carry higher levels of obligations and significance than statements of general support and are required for acceptance into AQ and IS (Blair et al., 2022; Berlin, 2024). We focus on pledges from organizations rather than pledges from individuals or small cells created by AQ or IS members in different countries. In our data, pledges come primarily from pre-existing unaffiliated jihadist organizations and jihadist splinter groups. This focus helps to isolate the disruptive impact of pledges to AQ/IS. Importantly, the formation of splinter groups that pledged to AQ or IS is associated with traditional measures of competition that focus on the number of groups in a conflict zone. However, organizations such as the Allied Democratic Forces, al-Shabaab, Boko Haram, and the Salafist Group for Preaching and Combat operated for years prior to pledging.

Arabic-language media and jihadist publications helped distinguish between different types of connections among jihadist actors. We supplemented these sources with scholarly publications, policy reports, and media articles. We recorded the year of the first pledge made by each group and the group’s main area of operations. Figure 1 illustrates the geographic distribution of these pledges. The map also shows that many countries with active armed groups lack such pledges, enabling empirical comparisons between contexts with and without transnational jihadist competition.

5.3 Ideology and Terrorist Designation

Religious ideology is a key dimension for capturing ideological distance between non-AQ/IS-affiliated groups and their transnational jihadist competitors. Using the EDTG, we code *religious group ideology* as a binary variable, classifying groups as either religiously motivated or not. While this approach does not capture more fine-

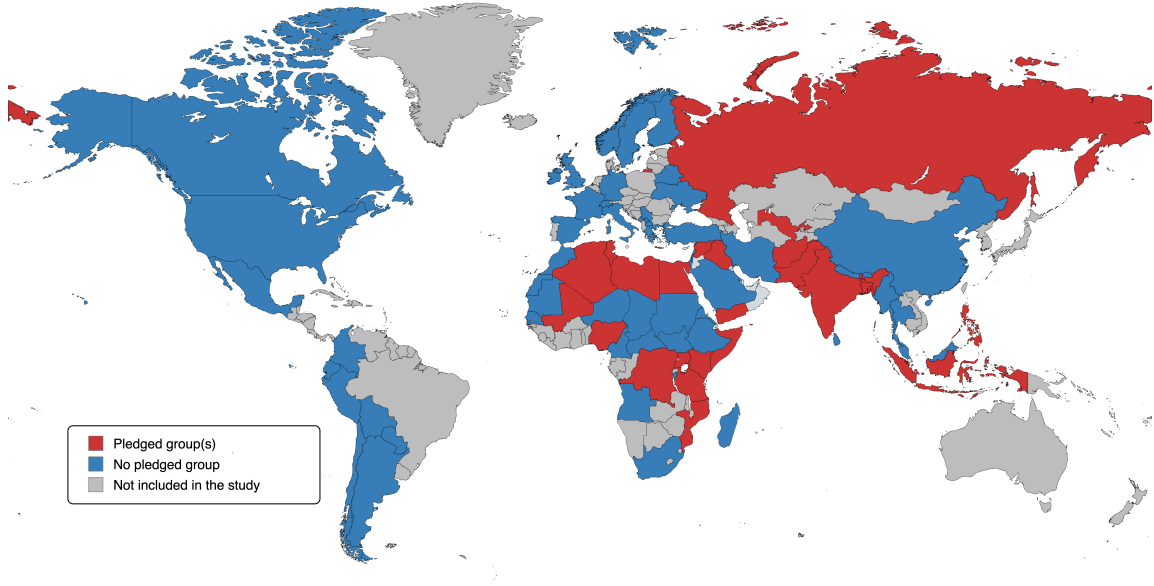


Figure 1: Geographic Distribution of Pledges to AQ and IS, 2004-2016

grained differences or ideological shifts over time (Ying, 2026), it remains the most feasible given our dataset’s cross-national scope.⁵

We measure *terrorist designation* using the U.S. Department of State’s Foreign Terrorist Organization (FTO) list,⁶ coding groups as designated or not. We follow recent scholarship analyzing the effects of terrorist proscription on militant behavior by focusing on U.S. designations of terrorist groups (Phillips, 2019). We do so given that U.S. designations carry significant weight in the international system and can influence the policies of other states (El Masri and Phillips, 2024).

⁵The EDTG classifies a group as religious if it espouses a religious fundamentalist ideology, excluding those that invoke religion solely as part of ethnic identity. Although the measure includes non-Islamist religious groups, only six appear in our 2004–2016 sample. As a robustness check, we recoded these groups as ideologically distant from AQ and IS affiliates and reran our subsample analyses. As shown in Appendix 13, the results remain substantively unchanged.

⁶See: <https://www.state.gov/foreign-terrorist-organizations/>.

5.4 Difference-in-Differences Design

We use a DiD design to assess how pledges to AQ or IS affect the civilian victimization behavior of non-AQ/IS-affiliated groups operating in the same country. Our analysis is conducted at the group-year level. Pledges serve as the treatment. We compare average changes in civilian victimization over time among treated groups (those in countries with pledges) and control groups (those in countries without pledges). While groups in pledge-affected countries may differ systematically from those in unaffected countries, the DiD design allows us to identify within-group changes over time, controlling for time-invariant group-level characteristics and time-variant confounders that apply uniformly across groups. Figure 2 displays the geographic distribution of treated groups.

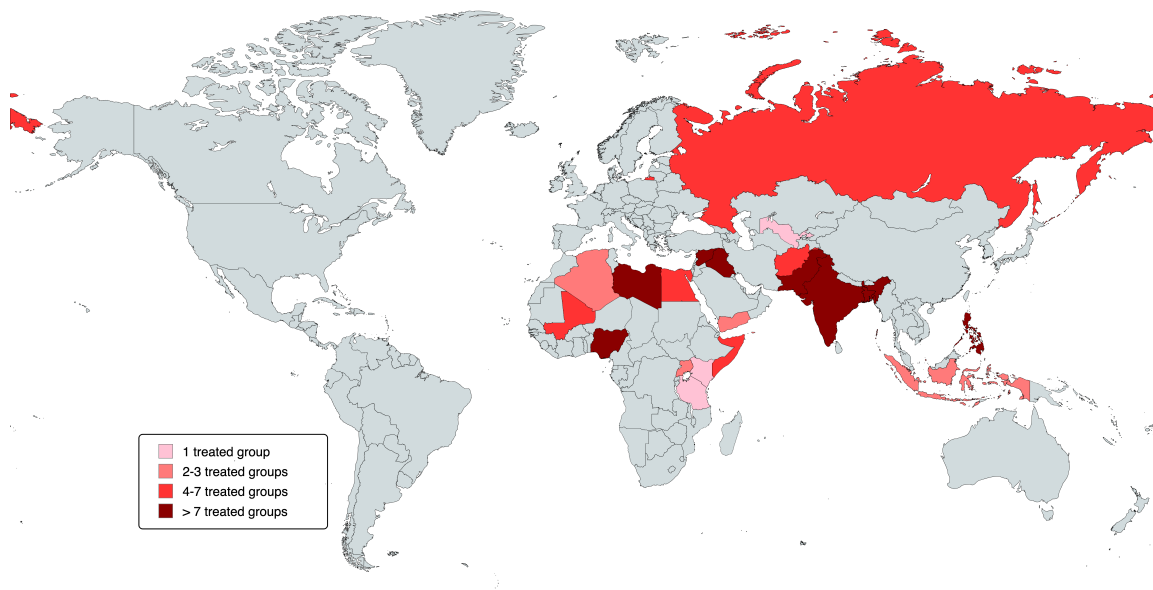


Figure 2: Geographic Distribution of Treated Groups, 2004-2016

Because pledges occur in different years, we use the staggered DiD framework developed by Callaway and Sant’Anna (2021), which is designed for heterogeneous

treatment timing. Unlike conventional two-way fixed effects (TWFE) estimators, which can be biased in staggered settings, this approach estimates cohort-specific effects relative to *not-yet-treated* or *never treated* controls and then aggregates them into group-time average treatment effects and overall averages, while also incorporating time- and unit-fixed effects. Our implementation considers both *not-yet-treated* or *never treated* units as controls, so comparisons are based on groups that have not yet been exposed to pledges at a given time.

Intuitively, for each cohort first treated at time t , the outcomes of treated groups after treatment are compared to the outcomes for groups untreated at that time; these cohort-time effects are then combined to summarize the causal impact over time. Figure 3 shows the annual number of newly treated groups, with 2014—coinciding with IS’s rise—marking the highest number of new treated groups (e.g., groups in countries with a pledge to AQ or IS).

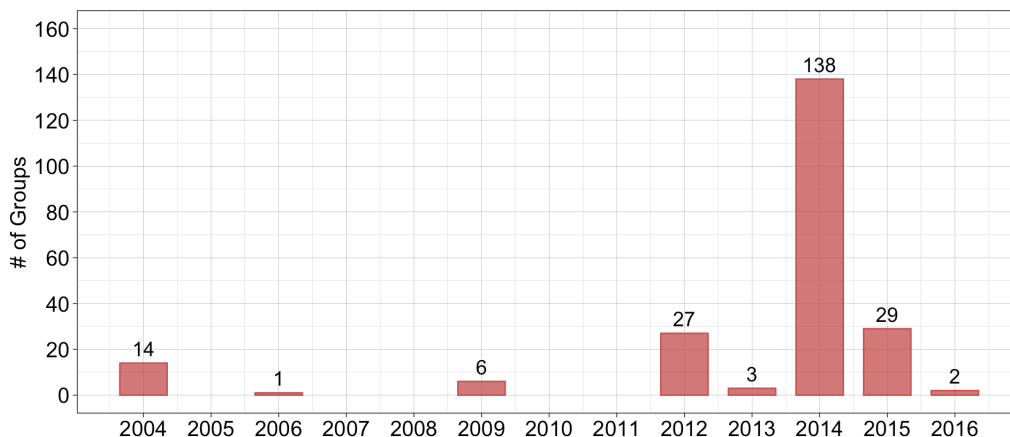


Figure 3: Number of Groups Treated Each Year, 2004-2016

In the extended models, we add time-varying controls: the group’s share of transnational attacks, attack diversity, binary indicators for kidnapping/extortion or leadership decapitation, group age. Details of these controls are in Appendix 1.

5.5 Parallel Trends

A core assumption of the DiD strategy is parallel trends: absent treatment (i.e., pledges), the average change in civilian victimization among treated groups would have mirrored that of control groups. If this holds, the model can isolate the treatment effect on outcome from confounders that influence both pledging and civilian targeting. This assumption might be compromised if country-level confounders affect both the likelihood of pledging and the manner in which unpledged non-AQ/IS-affiliated groups victimize civilians. However, the assumption is likely to hold in our context since prior research has identified leader- and group-level dynamics, such as organizational weaknesses and leadership deficits, as key drivers of pledging (Byman, 2014; Mendelsohn, 2015; Bacon, 2018; Blair and Potter, 2023).

A potential concern is that pledges are endogenously driven by domestic militant competition over resources, which might also influence civilian victimization. However, many jihadist groups operating in highly competitive environments, such as Ajnad Misr, Islamic Jihad Union, and the Moro Islamic Liberation Front, never pledged to AQ or IS. Others, like the Islamic Movement of Uzbekistan, pledged years after competition with rivals began (Berlin, 2024). Although this concern surrounding competition warrants consideration, it does not undermine the validity of our estimates. If pledges were primarily a response to escalating domestic competition, and if such competition increased civilian targeting by unpledged groups in the same country, we would expect pledges to correlate with higher levels of civilian victimization by unpledged non-AQ/IS groups. Our results, however, show the opposite pattern.

To further assess the parallel trends assumption, we empirically examine pre-

treatment trends.⁷ First, we compare civilian victimization rates between treated and control groups before the treatment. Parallel trends plots in Appendix 3 show that the trends were approximately parallel during this period. Second, following Auer’s (2003) suggestion, we conduct a pre-treatment regression analysis by interacting the DiD treatment variable with time dummies to create *leads* (e.g., dummy variables for pre-treatment periods) and *lags* (e.g., dummy variables for post-treatment periods). If the leads are statistically insignificant, it suggests that the treated and control groups were evolving in their civilian victimization behavior in parallel prior to treatment. If parallel trends hold, the interaction terms for the *leads* should be statistically insignificant, a pattern we observe.

Finally, we use event-study models to further probe pre-treatment imbalances (Hassell and Holbein, 2025). As shown in Figures 4–6, we find no statistically significant differences between the treatment and control groups, lending additional support to the parallel trends assumption.

5.6 Testing for Heterogeneous Treatment Effects

We extend our DiD analysis to test Hypotheses 3 and 4, which assess whether the effects of transnational jihadist competition vary by unpledged groups’ ideological orientation and prior terrorist designation. To do so, we apply the original staggered DiD model to the relevant subpopulations. For Hypothesis 3, we divide the sample into religious and non-religious unpledged groups and estimate the DiD model separately for each. For Hypothesis 4, we classify unpledged groups by their terrorist designation status prior to treatment and again estimate the model within each subset.

⁷While standard methods exist for evaluating parallel trends, none can definitively confirm it. These tests offer suggestive, not conclusive, evidence (Huntington-Klein 2021).

6 Results

We estimate the average treatment effects in our DiD setup, with *Pledges* serving as the treatment and *Civilian Victimization* by unpledged non-AQ/IS-affiliated groups (measured as the logged number of attacks against civilians) being the dependent variable. We use clustered bootstrapped standard errors at the group level.

6.1 Baseline Results

Our first set of results evaluates the effect of pledges to AQ or IS on the civilian victimization strategies of unpledged armed groups operating in the same country. Table 1 presents the weighted average of group-time average treatment effects. Model 1 provides a naive estimate of the treatment effect, while Model 2 introduces controls for targeting strategies that may influence victimization patterns. Model 3 accounts for leader decapitation and group age.

Across all specifications, non-AQ/IS-affiliated groups exhibit restraint toward civilians following pledges in their country of operation, contrasting with the expectations of outbidding (H1). The Average Treatment Effect on the Treated (ATT) of pledges on civilian victimization by non-AQ/IS-affiliated groups is consistently negative, though only statistically significant at the 90% confidence level across models. Estimates from the fully specified Model 3 suggest that pledges caused a 17.81% decrease in the expected number of attacks targeting civilians. However, the significance of the results at the 90% confidence level warrants a cautious interpretation, offering moderate, yet not definitive, support for the restraint hypothesis (H2) when applying DiD models to the entire sample without accounting for potential treatment heterogeneity by unpledged group’s ideology or designation status.

Further insights emerge from the event study analysis reported in Figure 4. Event

Table 1: Pledges and non-AQ/IS-Affiliated Groups' Civilian Victimization

	<i>DV: Civilian Victimization</i> All non-AQ/IS-Affiliated Groups		
	Model 1	Model 2	Model 3
Average Treatment Effect on the Treated	-0.1754* (0.0965)	-0.1599* (0.0947)	-0.1781* (0.0952)
Time-Fixed Effects	Yes	Yes	Yes
Unit-Fixed Effects	Yes	Yes	Yes
<i>Controls</i>			
Share of Transnational Attacks	No	Yes	Yes
Diversity of Attacks	No	Yes	Yes
Kidnapping/Extortion	No	Yes	Yes
Leader Decapitation	No	No	Yes
Group Age (logged)	No	No	Yes
N	2366	2301	2301

Note 1: **p < 0.05, *p < 0.1

Note 2: Robust standard errors clustered on group are in parentheses. All models in Tables 1, 2, and 3 are estimated with Callaway and Sant'Anna's (2021) doubly robust estimation method for staggered Difference-in-Differences using the `att_gt` function of the `did` package in R. All models consider both "Never Treated" and "Not Yet Treated" units as the control group.

time refers to the years before and after the announcement of pledges. Negative values of event time denote pre-treatment periods, whereas positive values denote the post-treatment period. Dynamic group-time average treatment effects⁸ reveal no significant differences between treatment and control units in the pre-treatment period, confirming parallel trends. Post-treatment estimates show a downward shift in civilian victimization by unpledged groups following pledges in their country of operation. While some estimates achieve only 90% confidence (red markers), others reach 95% (blue markers).

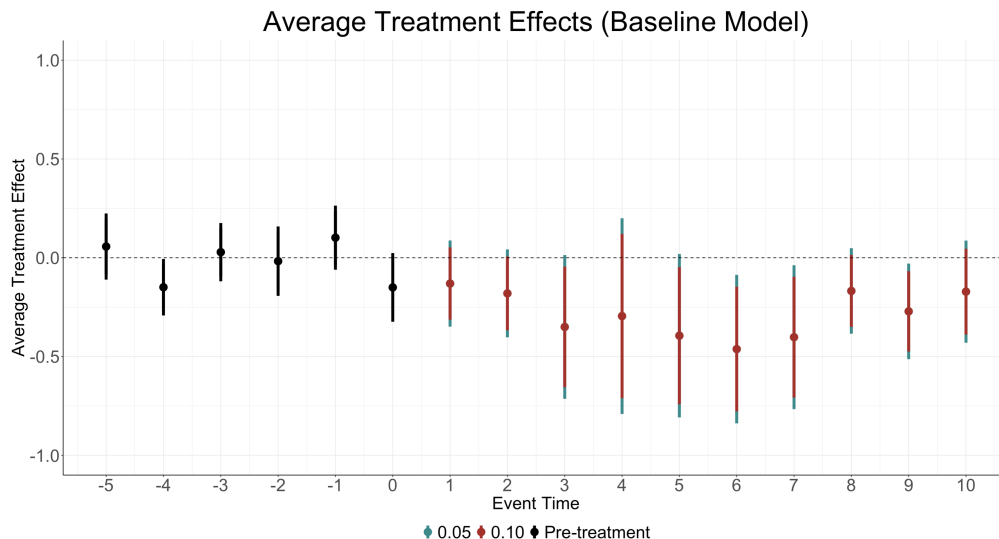


Figure 4: Aggregate Group-Time Average Treatment Effects on the Treated (Event Study), 2004-2016

Note: Event study is based on Model 1 of Table 1. Event time refers to the years before and after the announcement of pledges. Dynamic group-time average treatment effects are computed using the `aggte` function of the `did` package in R.

⁸This procedure computes average effects across varying lengths of treatment exposure and mirrors an event study.

6.2 Heterogeneous Treatment Effects by Ideology

We next examine the effects of pledges on civilian victimization by non-AQ/IS-affiliated groups by disaggregating these groups based on ideology—religious versus non-religious—to evaluate whether ideological distance with transnational jihadists moderates non-AQ/IS-affiliated groups’ responses to competition (H3). Separate DiD models are run for unpledged religious ($N = 136$) and non-religious groups ($N = 349$). Table 2 presents these results. Religious unpledged groups’ ATTs (Models 4–6) are consistently negative, albeit statistically insignificant. Importantly, while these results do not support the restraint hypothesis, neither do they yield evidence consistent with the conventional outbidding hypothesis. Religious unpledged groups do not seem to escalate (or significantly restrain) civilian targeting in response to competition with AQ and IS affiliates.

Table 2: Pledges and Local Groups’ Civilian Victimization by Group Ideology

	<i>DV: Civilian Victimization</i>					
	Sample: Religious Groups			Sample: Non-Religious		
	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
ATT	-0.176 (0.1895)	-0.123 (0.226)	-0.084 (0.1431)	-0.19** (0.0959)	-0.180* (0.1058)	-0.192* (0.1057)
Time-Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Unit-Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
<i>Controls</i>						
Share of Transnational Attacks	No	Yes	Yes	No	Yes	Yes
Diversity of Attacks	No	Yes	Yes	No	Yes	Yes
Kidnapping/Extortion	No	Yes	Yes	No	Yes	Yes
Leader Decapitation	No	No	No	No	No	Yes
Group Age (logged)	No	No	No	No	No	Yes
N	624	585	585	1742	1716	1716

Note: ** $p < 0.05$, * $p < 0.1$. Robust standard errors clustered on group are in parentheses.

In contrast, non-religious unpledged groups (Models 7–9) show stronger support

for both the overall restraint hypothesis (H2) and the *ideological distance* hypothesis (H3). The ATTs are negative and significant at the 95% confidence level in Model 7 and the 90% level in Models 8 and 9. Although the 90% threshold reflects a modest level of statistical significance, the consistency and direction of the effects in all models provide suggestive evidence that ideological distance from transnational jihadist actors conditions how unpledged groups adapt their strategies. Estimates from the fully specified Model 9 indicate pledges reduced non-religious, non-AQ/IS-affiliated groups' attacks against civilians by 19.17%.

Figure 5 extends this analysis through event studies using Models 4 and 7. The left panel shows religious unpledged groups maintaining parallel trends pre-treatment, but post-treatment declines fail to achieve statistical significance, even at the 90% level. In contrast, the right panel illustrates stronger patterns for non-religious unpledged groups (72 percent of the groups we analyze). Pre-treatment effects hover around zero, supporting parallel trends. Post-treatment effects reveal sustained reductions in civilian targeting, achieving 95% significance (blue markers) starting from year 3 onward. These results provide support for H3, which posits ideological distance as a key factor in shaping unpledged local groups' responses to transnational jihadist competition.

6.3 Heterogeneous Treatment Effects by Terrorist Designation

We further test whether the effects of pledges depend on non-AQ/IS-affiliated groups' terrorist designation status (H4) in Table 3. Specifically, we hypothesize that non-designated unpledged groups will moderate violence more than designated unpledged groups. Separate DiD models analyze designated ($N = 37$) and non-designated unpledged groups ($N = 448$).

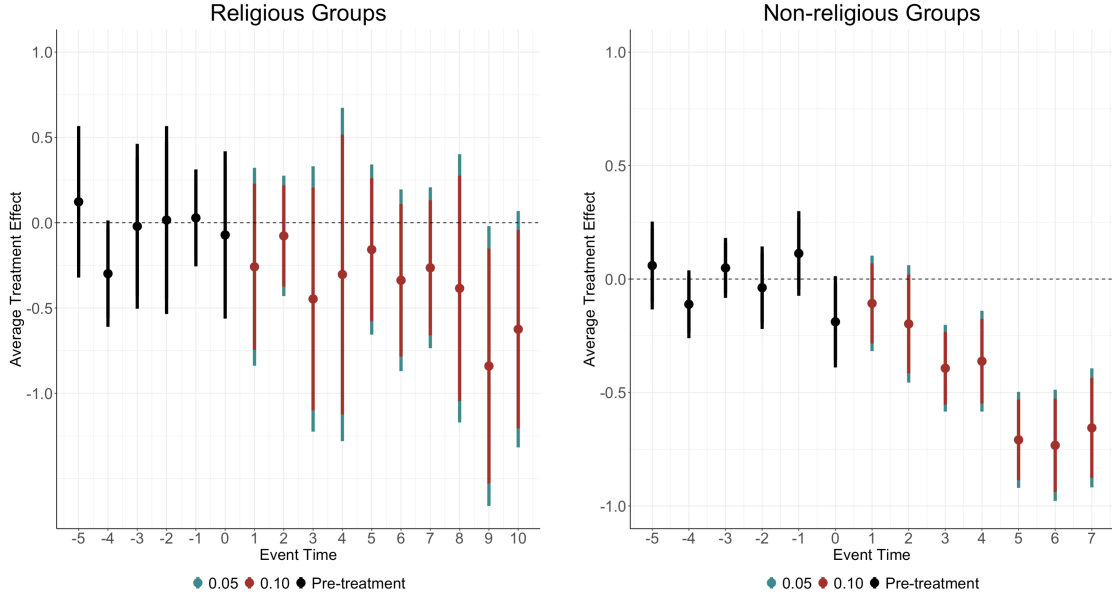


Figure 5: Aggregate Group-Time Average Treatment Effects on the Treated by Group Ideology

Note: Event study is based on Models 4 and 7 of Table 2. Dynamic group-time average treatment effects are computed using the `aggte` function of the `did` package in R.

Among designated unpledged groups (Models 10–12), ATTs are negative but statistically insignificant. This aligns with expectations, as designated groups face limited prospects for reputational improvement. By contrast, non-designated unpledged groups (Models 13–15) exhibit reductions in civilian victimization, supporting H4. The ATTs are negative and significant at conventional levels across specifications, with Model 15 estimating a 23% decrease in civilian targeting by non-designated, non-AQ/IS-affiliated groups following a pledge in their country.

The event study results in Figure 6 that visualize the dynamic effects associated with our models corroborate these findings. For designated unpledged groups (left panel), post-treatment effects remain insignificant, reinforcing the lack of moderation. In contrast, non-designated unpledged groups (right panel) exhibit significant

Table 3: Pledges and Local Groups’ Civilian Victimization by Designation Status

	<i>DV: Civilian Victimization</i>					
	Sample: Designated Groups			Sample: Non-Designated		
	Model 10	Model 11	Model 12	Model 13	Model 14	Model 15
ATT	-0.361 (0.36)	-0.008 (0.4146)	-0.531 (0.5953)	-0.19** (0.0911)	-0.21** (0.1019)	-0.23** (0.1149)
Time-Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Unit-Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
<i>Controls</i>						
Share of Transnational Attacks	No	Yes	Yes	No	Yes	Yes
Diversity of Attacks	No	Yes	Yes	No	Yes	Yes
Kidnapping/Extortion	No	Yes	Yes	No	Yes	Yes
Leader Decapitation	No	No	No	No	No	Yes
Group Age (logged)	No	No	No	No	No	Yes
N	234	234	234	2067	2002	2002

Note: **p < 0.05, *p < 0.1. Robust standard errors clustered on group are in parentheses.

and sustained reductions, reaching 95% confidence (blue markers) by years 5–7 and again in year 9. Non-designated groups—which have incentives to avoid blacklisting and greater prospects for improving their reputations—demonstrate restraint toward civilians following pledges in their country of operation, potentially leveraging competition with transnational jihadists to enhance their international appeal.

Importantly, the temporal dynamics across different findings highlight that the adoption of restraint does not occur immediately. Our coding strategy likely attenuates short-term effects because pledges are measured at the calendar-year level, which biases against finding very short-term effects. On the other hand, one potential theoretical explanation for these temporal dynamics is that armed groups develop expertise in particular forms of violence and executing specific operational tasks. While such routines facilitate the implementation of established behaviors, they can also constrain adaptation by locking organizations into particular “ways of

doing business” (Horowitz, 2010, 39). Armed groups that previously relied heavily on civilian victimization may, consequently, require time to recalibrate their violence and targeting practices as organizational members move away from established operational activity.

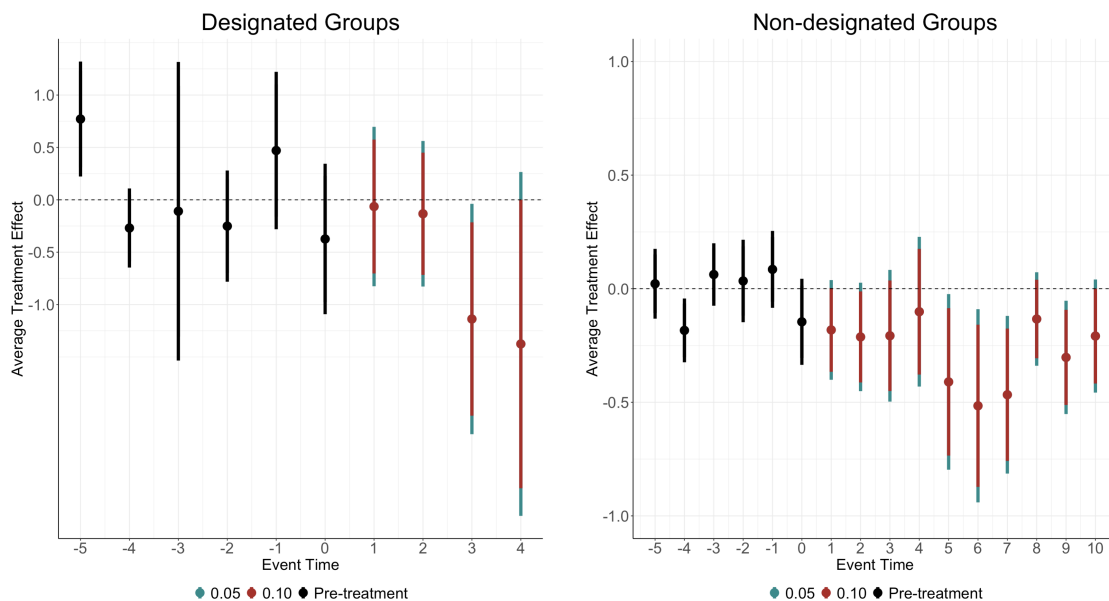


Figure 6: Aggregate Group-Time Average Treatment Effects on the Treated by Designation Status

Note: Event study is based on Models 10 and 13 of Table 3. Dynamic group-time average treatment effects are computed using the `aggte` function of the `did` package in R.

7 Robustness and Extensions

7.1 Alternative Measures

We estimate a series of negative binomial regression models as robustness checks. As shown in Appendix 12, the results remain consistent. We also use an alternative

measure of civilian victimization: *logged number of civilian casualties*, with the results (Appendix 4) closely mirroring our primary analysis. Finally, we run the analyses using a version of the outcome measure based on a much narrower definition of civilian targets (Appendix 16).

Next, we employ an alternative measure of group ideology using data collected by Piazza (2009) and extended by Hand and Saiya (2022) on “strategic” and “universal/abstract” armed groups. While strategic groups have limited and concrete objectives, such as liberating a specific territory, universalist groups espouse “highly ambitious, abstract, complex, and nebulous goals that are driven primarily by ideology” (Piazza, 2009, 65). Using this proxy, we replicate our main findings. Pledges in their country of operation lead strategic unpledged groups to reduce civilian victimization, while universalist unpledged groups, a category that includes numerous religious organizations, do not (Appendix 4).

Finally, to address concerns that pledging groups formed through splintering may affect competition differently by adding more groups to a conflict zone, we re-estimate all models after excluding pledges that were issued by recently formed splinters from our treatment. The results remain substantively unchanged (Appendix 14).

7.2 Target Specificity of Restraint

To assess whether the observed decline in civilian targeting simply reflects a broader reduction in overall armed activity due to capacity constraints, we examine attacks against police and military targets (Appendix 15). Once group-level controls are included, we find no consistent evidence of decreased violence against these targets following pledges to AQ and IS. These patterns suggest that the observed restraint toward civilians is not simply a by-product of declining activity or capacity.

7.3 Resource-Centric Concerns

Although our DiD models capture within-group changes over time, we also address potential confounding from time-varying group-level factors. We control for two factors that plausibly shape civilian victimization—*foreign state support* (a proxy of reduced dependence on local populations) and *social service provision* (a proxy of greater concern for local populations). Both measures are derived from the EDTG dataset. The results (Appendix 5 and 6) remain consistent with our main findings, indicating that observed restraint is not driven by changes in unpledged groups’ resources or population-oriented strategies.

7.4 Militant Infighting

Militant infighting—violence between armed groups—can induce restraint by intensifying resource pressures and increasing groups’ dependence on civilians (Polo and Welsh, 2024). If non-AQ/IS-affiliated groups frequently clash with AQ/IS affiliates, observed restraint might reflect resource reallocation to infighting rather than AQ/IS affiliates shifting unpledged groups’ incentives for restraint. To account for this, we control for the propensity of unpledged groups to attack other armed groups, using the GTD’s target-type indicators. The results (Appendix 11) remain consistent. We also re-estimate our models using only group-years with minimal infighting. Even in these cases, unpledged non-religious and non-designated groups continue to exhibit restraint following pledges in their country of operation.

7.5 Organizational Capacity and Territorial Aspirations

We further test robustness by controlling for two group-level factors related to civilian targeting: non-AQ/IS affiliated groups’ *organizational strength* and *secessionist goals*.

Both measures are derived from the EDTG. Weaker groups may be unable to escalate violence if AQ/IS-affiliated rivals outcompete them for recruits, funding, or territory, which raises the possibility that such unpledged groups' restraint stems from limited capacity rather than deliberate moderation. Likewise, secessionist unpledged groups may be especially vulnerable to competition from AQ/IS affiliates, who could directly threaten their territorial ambitions. After controlling for both factors, our results (Appendix 7 and 8) remain unchanged.

7.6 Transnational Jihadist Violence

Transnational jihadists' use of extreme violence can serve as a competitive reference point that raises the costs of engaging in violent outbidding. To account for this, we use the GTD to calculate the *logged number of civilian casualties caused by pledged AQ/IS-affiliated groups* per country-year. We merge this measure with our dataset, linking it to unpledged groups operating in the same country, and include it as a control in our DiD models. The results (Appendix 9) remain consistent.

7.7 Spatial Substitution

We also consider the possibility of spatial substitution: unpledged groups might shift operations across borders to avoid competition with transnational jihadists. If so, the observed decline in their civilian victimization could reflect geographic displacement rather than restraint. To mitigate this, we re-estimate our DiD models using the *logged number of cross-border civilian attacks* as the outcome (Appendix 10). The consistency of our main findings strengthens the argument that the observed effects are not driven by geographic displacement.

8 Heterogeneity by Transnational Affiliate

Our baseline analysis treats pledges to al-Qaeda and the Islamic State as equivalent treatments, focusing on the broader competitive effects of competition with transnational jihadist organizations. However, the incentives for unpledged groups to use restraint to behaviorally distance themselves from highly stigmatized jihadist actors may be particularly pronounced when competing with jihadist organizations that pledged allegiance to IS. In the last decade, IS has attracted exceptional international stigma through its extreme violence, declaration of a transnational caliphate, and prominence in global counterterrorism policy and media coverage (Friis, 2018). As mentioned above, despite its history of mass casualty attacks and use of indiscriminate violence, AQ has increasingly emphasized moderation in recent years, especially in attempts to differentiate itself from IS (Hamming, 2020). Reputational and competitive pressures may, consequently, be especially strong following the emergence of IS affiliates.

Estimating these effects separately is empirically challenging. AQ and IS pledges frequently co-occur within the same countries: 14 of the 15 countries experiencing an AQ pledge also experience an IS pledge, and 14 of the 23 countries with an IS pledge also have an AQ pledge. One primary reason for this is that various mid-ranking commanders of AQ affiliates splintered to pledge allegiance to IS and create their own organizations (Warner et al., 2022). Disaggregating the analysis therefore substantially reduces statistical power. With this caveat in mind, we estimated separate models for each organization by subsetting countries that experienced only one type of pledge. As reported in Appendix Tables A.24–A.29, the models isolating the effects of IS pledges produce estimates substantively similar to the main analysis, while models isolating AQ pledges yield null results. Although noting that the statistical

significance is attenuated in our “IS Pledge“ models (Appendix Tables A24-A.26) compared to the main models, the pattern supports the idea that pledges to IS are the primary driver of the behavioral shifts we document among unpledged armed groups operating in the same country as a pledged group.

9 Conclusion

This study examined the strategic behavior of armed groups competing with transnational jihadist organizations. Contrary to expectations surrounding outbidding, we find that competition with transnational jihadist actors is associated with a moderate decline in civilian victimization. However, these findings are primarily driven by a certain subset of armed actors, indicating that not all armed groups respond to competition with AQ and IS with restraint. Overall, these findings highlight that a) the type of rival involved in a conflict can affect the behavior of other armed groups and b) not all organizations respond to competition in the same way.

Future research could build on this study in several ways. First, scholars have investigated how violence by armed groups affects their legitimacy and support among external audiences (Arves, Cunningham and McCulloch, 2019). Additional studies could examine whether the use of restraint, in isolation or in combination with other tactics, increases support among external populations, especially when compared with the behavior of more extreme actors. Moreover, armed groups may combine restraint against civilians with changes in social service provision, discourse, and ideological beliefs to differentiate themselves from rivals in fragmented conflicts. Additional scholarship could investigate whether these constraining measures affect the long-term preferences and goals of armed groups (Loyle et al., 2023).

There are also numerous avenues for advancing scholarship on militant competi-

tion. Additional research could continue to analyze the behavior of religious groups during competition (Hamming, 2020; Berlin and Rangazas, 2023; Schwab, 2023). While our findings show that religious groups did not significantly moderate in response to pledges to AQ or IS, neither did they escalate civilian targeting. This challenges findings that religious groups are indifferent to public judgment and prone to violent escalation during competition (Nemeth, 2014). Moreover, our findings indicated that the adoption of restraint takes time, suggesting the need for further investigation surrounding the temporal dynamics surrounding outbidding and restraint. Finally, this article highlighted the need to consider the *type* of actor engaged in competition. Future studies could investigate whether other types of armed actors, such as international criminal organizations, which operate in gray zones between political and economic violence, may shape competition in ways that scholars have not yet considered. Understanding these dynamics remains a pressing concern due to the high number of fragmented civil wars in the international system.

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